PMCC & Non-linear Regression

Exam code: 9MA0

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Product moment correlation coefficient (PMCC)

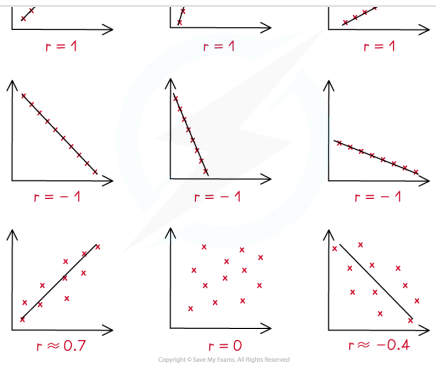
What is the product moment correlation coefficient?

- The product moment correlation coefficient (PMCC) is a way of giving a numerical value to **linear correlation** of bivariate data
- The PMCC of a sample is denoted by the letter r
  - r can take any value such that $-1 \leq r \leq 1$
  - A positive value of r describes positive correlation
  - A negative value of r describes negative correlation
  - If $r = 0$ there is no correlation
  - $r = 1$ means perfect positive correlation and $r = -1$ means perfect negative correlation
  - The closer to 1 or -1, the stronger the correlation
- The gradient does not change the value of r

Stuck on something?

Ask questions and get help with exactly what you're revising.

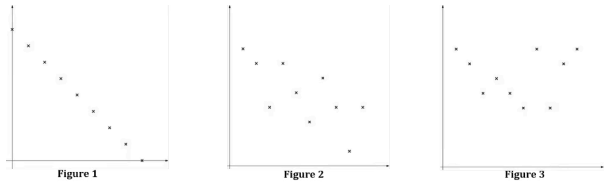
Got it



How is the product moment correlation coefficient calculated?

- You must learn how to use your calculator to calculate value of the PMCC, r for the relationship between two variables
- All calculators are different and you should make sure you can calculate the PMCC on your personal calculator
  - Make sure you know how put your calculator into the statistics mode
    - You will be given the option to turn the frequency on or off, choose off for most calculations of the PMCC
  - With the statistics mode switched on on your calculator, there will be a 'statistics' option, followed by a regression option in the form A + BX
    - Your calculator will give you two columns into which you can input the x and y data values
  - Once the data has been entered into your calculator, choose the 'r' value from the 'STAT' options

Worked Example



Three scatter diagrams, showing observations from different bivariate data sets, are shown above.

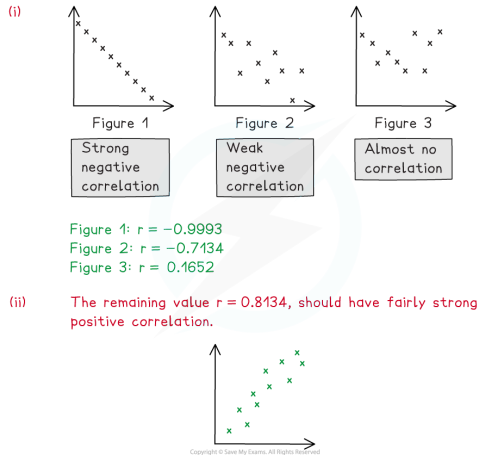
(i) Match each of the three scatter diagrams show above to one of the values of r given below. You should use each given value of r no more than once.

Answer:

$$r = -0.9993$$

(ii) Sketch a scatter diagram for the remaining value of r listed above.

Answer:



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Non-linear regression

You have learned how to use linear regression models to describe a relationship between two variables. However, it is possible for two variables to have a relationship that does not fit a linear model, but still shows a pattern based on exponential growth or decay. A linear regression model is only appropriate if the PMCC is close to 1 or -1.

What are non-linear regression models?

- If a **bivariate data set** appears to have a non – linear relationship it could fit an exponential model
  - A non – linear regression model could take the form $y = ax^n$ or $y = kb^x$ where a, n, k and b are constants

- If the regression model takes the form $y = ax^n$ the data should be coded from x - values to y - values using $X = \log x$ and $Y = \log y$
  - If $y = ax^n$ for constants *a* and *n*, then $\log y = \log a + n\log x$ or $Y = nX + \log a$
  - Plotting $\log x$ against $\log y$ will give a linear graph
  - The y – intercept would be $\log a$ and the gradient of the line would be *n*
  - This can be shown by taking logarithms of both sides

- If the regression model takes the form $y = kb^x$ the data should be coded from x values to y values using $X = x$ and $Y = \log y$
  - If $y = kb^x$ for constants *k* and *b* , then $\log y = \log k + x\log b$ or $Y = (\log b)X + \log k$
  - Plotting x against $\log y$ will give a linear graph
  - The y – intercept would be $\log k$ and the gradient of the line would be $\log b$
  - This can be shown in the same way by taking logarithms of both sides
  - For example:

$$y = kb^x$$

Take logarithms of both sides

$$\log y = \log(kb^x)$$

Use the addition law for logarithms

$$\log y = \log k + \log b^x$$

Use the power law for logarithms

$$\log y = \log k + x \log b$$

- Using logarithms to code the data in this way is called **changing the variables**

How do I use non–linear regression models?

- Non – linear regression models can be used in much the same way as **linear regression models**
- By coding the original data using logarithms (changing the variables) a regression line of *Y* on *X* can be found
  - This can be used to make predictions for data values that are within the range of the given data (**interpolation)**
  - Making a prediction outside of the range of the given data is called **extrapolation** and should not be done

- The non – linear regression model can then be found by substituting $\log x$ and $\log y$ back into the *X* and *Y* values in the regression line and rearranging

Worked Example

using the form $t = k\,h^n$.

The data are coded using the changes of variables $X = \log h$ and $Y = \log t$. The regression line of Y on X is found to be $Y = -3.5X$.

(i) Find the values of X and Y for a child that is 75 cm tall and naps for 4 hours per day, giving your answers to four decimal places.

(ii) Using the regression line, show that a child of height 0.9 metres would be expected to nap for approximately 1.45 hours per day.

(iii) State an assumption that was made in order to justify the use of the regression line in part (ii).

(iv) By first substituting $\log h$ for X and $\log t$ for Y in the equation of the regression line given, show that the relationship between the height of a child and the time they spend sleeping can be modelled by $t = h^{-3.5}$.

Answer:

Examiner Tips and Tricks

- Be careful when using original and coded data interchangeably, it is easy to forget which one you are working with. Remember that if your regression line was calculated using coded data then you will need to reverse this if finding predictions. Make sure that you are familiar with using logarithms, indices and their laws. Be careful to check which base logarithms were used for coding the data, if $\log x$ was used then it is reversed using $10^{\log x}$, but is $\ln x$ was used then it should be reversed using $e^{\ln x}$.

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